



# CIE A Level Chemistry



Your notes

## 9.3 Chemical Periodicity of Other Elements

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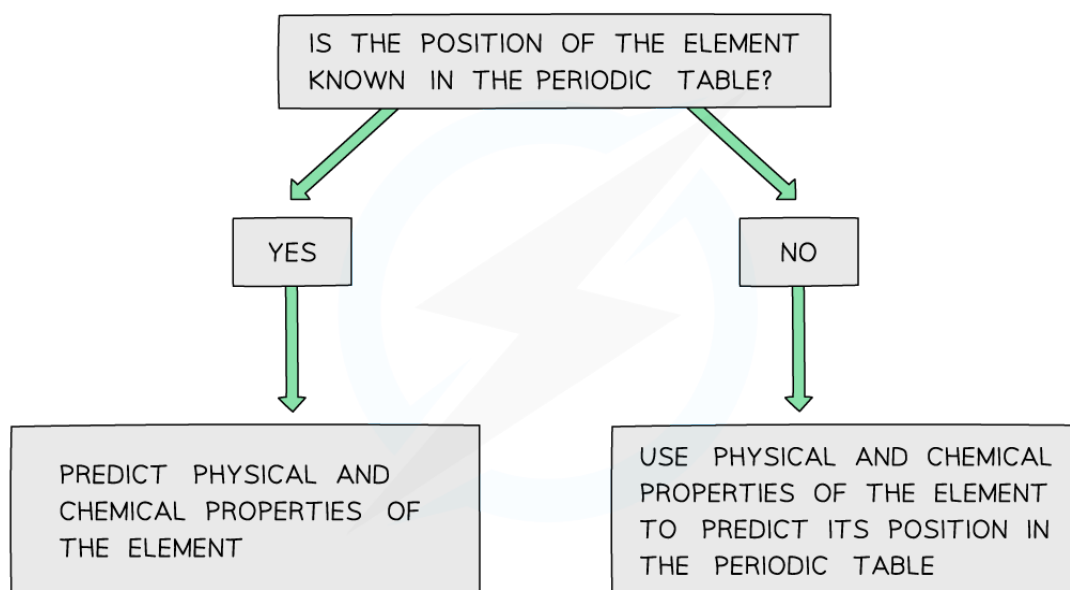
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## Chemical Periodicity of Other Elements

### Periodicity: Predicting Position & Properties

- If the **chemical** and **physical properties** of an element are known, the **position** of that element in the Periodic Table can be predicted
- Similarly, predictions can be made about the **physical** and **chemical properties** of elements if the **position** of the element in the Periodic Table is known

The diagram shows a flowchart to make predictions about an element's behaviour



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*The flowchart shows how to use the position and / or properties of an element to make predictions about its behaviour*



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 **Worked example****Deducing the position of an element in the Periodic Table**

Element **X** forms a chloride which reacts with water to form a solution of pH 1.0.

It forms an oxide which has a melting point of 1610 °C.

The oxide does not dissolve in or react with aqueous sodium hydroxide.

Deduce the possible position of element **X** in the Periodic Table.

**Answer**

- Break the question down and systematically approach the question
- **Step 1:** 'Element **X** forms a chloride, which reacts with water to form a solution of pH 1'
  - The low pH of the solution formed suggests that the chloride is a non-metallic chloride (group 13 to 17)
- **Step 2:** 'The oxide does not dissolve in or react with aqueous sodium hydroxide'
  - Since aluminium oxide does react with sodium hydroxide, element **X** cannot be Group 13
- **Step 3:** 'It forms an oxide which has a melting point of 1610 °C'
  - This suggests a giant molecular (covalent) structure which corresponds to Group 14
- **Step 4:**
  - Element **X** cannot be carbon (which is in Group 14) as carbon dioxide is a gas whereas the element X oxide is a solid (with a melting point of 1610 °C)
- **Step 5:**
  - Element **X** is therefore a Group 14 element in Period 3 or lower
- Note that this is an example of predicting the position of an element based on its physical and chemical properties



Your notes

 **Worked example****Predicting physical and chemical properties of selenium**

Selenium is in Group 16 and Period 4 of the Periodic Table.

Predict some physical and chemical properties of selenium.

**Answer**

- **Step 1:** Selenium is a nonmetal in Group 16
  - If compared to sulfur, its structure is predicted to be a **simple molecular structure**
- **Step 2: Physical properties**
  - Simple molecular structures have **low melting points, do not conduct electricity** and are **insoluble** in water
- **Step 3: Chemical properties** are:
  - Reaction with **water**
    - Simple molecules do not react with water
  - Reaction with **oxygen**
    - Simple molecules react with oxygen
    - So, selenium could form an oxide of  $\text{SeO}_2$  (if compared with sulfur)
  - Reaction with **chlorine**
    - Simple molecules react with chlorine
    - So, selenium could form a simple molecule of  $\text{SeCl}_2$  which can react with the water vapour in the air to form hydrogen chloride (if compared with sulfur)
- Note that this is an example of predicting the chemical and physical properties of an element based on its position in the Periodic Table



Your notes

 **Worked example****Determining an unknown chloride compound**

The chloride of unknown element **Y** is a liquid at 20 °C.

This chloride reacts with water, releasing white fumes and leaving an acidic solution.

1. Does element **Y** belong to Group 1, Group 2 or Group 15 of the Periodic Table?
2. What type of reaction takes place between the chloride of **Y** and water?
3. Identify the white fumes given off when the chloride of **Y** reacts with water.

**Answer****Answer 1:**

- Group 1 and Group 2 chlorides react with water to form colourless solutions of pH 6.5 – 7.0
- Group 15 chlorides on the other hand react with water to form strong acidic solutions
- Element **Y** therefore belongs to Group 15 of the Periodic Table

**Answer 2:**

- Group 15 chlorides get hydrolysed in water, therefore this is a hydrolysis reaction

**Answer 3:**

- The white fumes that are given off during this reaction are hydrogen chloride gas